

Running Nextflow with built-in docker support

Results

We see a generated fastqc and multiqc reports in results folder



Running Nextflow with conda

```
(nextflow) egoncharov@tux:~$ nextflow run nextflow-io/rnaseq-nf -profile conda

NEXTFLOW ~ version 25.10.4
Launching 'https://github.com/nextflow-io/rnaseq-nf' [infallible_planck] DSL2 - revision: ca20a6dfd2 [master]

RNASEQ - NF PIPELINE
=====
transcriptome: /home/egoncharov/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads         : /home/egoncharov/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir        : results

executor > local (3)
[00/b70edc] process > RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✓
[ac/f80b20] process > RNASEQ:FASTQC (ggal_gut) [100%] 1 of 1 ✓
[cb/2daa72] process > RNASEQ:QUANT (ggal_gut) [100%] 1 of 1 ✓
[ ] process > MULTIQC -
Creating env using conda: bioconda::multiqc=1.27.1 [cache /home/egoncharov/work/conda/env-4a436c230263dfdbbf4ddd0623505d1]

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outdir        : results

executor > local (4)
[00/b70edc] process > RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✓
[ac/f80b20] process > RNASEQ:FASTQC (ggal_gut) [100%] 1 of 1 ✓
[cb/2daa72] process > RNASEQ:QUANT (ggal_gut) [100%] 1 of 1 ✓
[35/a5435f] process > MULTIQC [100%] 1 of 1 ✓

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 22:04:42
Duration    : 2m 45s
CPU hours   : (a few seconds)
Succeeded   : 4

(nextflow) egoncharov@tux:~$
```

Results

```
(nextflow) egoncharov@tux:~$ cd results
(nextflow) egoncharov@tux:~/results$ ls
fastqc_ggal_gut_logs multiqc_report.html
(nextflow) egoncharov@tux:~/results$ cd fastqc_ggal_gut_logs/
(nextflow) egoncharov@tux:~/results/fastqc_ggal_gut_logs$ ls
ggal_gut_1_fastqc.html ggal_gut_1_fastqc.zip ggal_gut_2_fastqc.html ggal_gut_2_fastqc.zip
(nextflow) egoncharov@tux:~/results/fastqc_ggal_gut_logs$
```

Results are the same

 multiqc 1.24

Nextflow RNA-seq demo

General Stats

Salmon

FastQC

Sequence Counts

Sequence Quality Histograms

Per Sequence Quality Scores

Per Base Sequence Content

Per Sequence GC Content

Per Base N Content

Sequence Length Distribution

Sequence Duplication Levels

Overrepresented sequences by sample

Top overrepresented sequences

Adapter Content

Status Checks

Software Versions

 multiqc

Nextflow RNA-seq demo

A pipeline for quantification of genomic features from short read data

A modular tool to aggregate results from bioinformatics analyses across many samples into a single report.

Contact E-mail: pablo.oliveros@gmail.com

Application Type: RNA-seq

Project Type: Nextflow demo

Report generated on 2026-04-09, 22:04 MDT based on data in: /home/egoncharov/work/75/c4543f/nextflow/731/fault4a23094

Welcome! Not sure where to start? [Watch a tutorial video](#) (3:00) [Start over again](#) (x)

General Statistics

Copy table

Configure columns

Scatter plot

Violin plot

Export as CSV

Showing % rows and % columns

Summarize table

Sample Name	% Aligned	M Aligned	CFR	M Bias	Dups	GC	Seqs
ggal_gut_1					16.3%	45.6%	8.0 M
ggal_gut_2					16.2%	45.5%	8.0 M
quant_ggal_gut	87.6%	8.0 M	100.0%	0.6			

Salmon Version: 1.10.3

Quantifies expression of transcripts using RNA-seq data. URL: <https://bioinformatics.github.io/salmon/> DOI: [10.1093/bioinformatics/bty119](https://doi.org/10.1093/bioinformatics/bty119)

Summarize plot

Export...

Running pipeline on my own data

I've already had some downloaded RNA-seq data (Lymphoblastoid transcriptomes from 1000 Genomes Geuvadis dataset)

So let's run our pipeline on it

- To speed up the process, i've cut the initial .fastq file to 100_000 reads

```
[base] egoncharov@ec2-~$ cat /data/bio/human_genomics/vaccine/external/geuvadis/trimmD/ERR204895_1.fastq.gz | head -n 400000 | gzip > /home/egoncharov/Temporary_Things/BioInf_Pipelines/HM1/fastq_data/ERR204895_subset_1.fastq.gz
[base] egoncharov@ec2-~$ cat /data/bio/human_genomics/vaccine/external/geuvadis/trimmD/ERR204895_2.fastq.gz | head -n 400000 | gzip > /home/egoncharov/Temporary_Things/BioInf_Pipelines/HM1/fastq_data/ERR204895_subset_2.fastq.gz
```

- Then, the standard pipeline with docker profile was run

```
nextflow run egoncharov@ec2-~$ nextflow run nextflow-to/rnaseq-nf --reads /home/egoncharov/Temporary_Things/BioInf_Pipelines/HM1/fastq_data/ERR204895_subset_{1,2}.fastq.gz --profile docker \
--transcriptome /data/bio/human_genomics/shared/maps/GDC.h38.d1.vd1/viruses/Gencode.v47/transcriptome/gencode.v47.chr_patch_hapl_scaff.annotation.dic.fa \
--outdir /home/egoncharov/Temporary_Things/BioInf_Pipelines/HM1/result_for_subset

nextflow --version 25.10.4
Launching https://github.com/nextflow-io/rnaseq-nf [thirsty_mirzakhanli] DSL2 - revision: ca20a6fd2 [master]

RNASEQ-NF PIPELINE
=====
transcriptome: /data/bio/human_genomics/shared/maps/GDC.h38.d1.vd1/viruses/Gencode.v47/transcriptome/gencode.v47.chr_patch_hapl_scaff.annotation.dic.fa
reads: /home/egoncharov/Temporary_Things/BioInf_Pipelines/HM1/fastq_data/ERR204895_subset_{1,2}.fastq.gz
outdir: /home/egoncharov/Temporary_Things/BioInf_Pipelines/HM1/result_for_subset

executor > local (4)
[ec2-472a80] process > RNASEQ:INDEX (gencode) [100%] 1 of 1 -
[ec2-472a80] process > RNASEQ:FASTQC (ERR204895_subset) [100%] 1 of 1 -
[ec2-81b6b0] process > RNASEQ:QUANT (ERR204895_subset) [100%] 1 of 1 -
[ec2-7f447d] process > MULTQC [100%] 1 of 1 -

Done! Open the following report in your browser --> /home/egoncharov/Temporary_Things/BioInf_Pipelines/HM1/result_for_subset/multiqc_report.html

Completed at: 09-Apr-2026 23:05:58
Duration: 11m 42s
CPU hours: 0.2
Successes: 4
```

Results

